

LUCAS MARIANO GRIGOLATO

DIGITAL DESIGN, EMBEDDED & TELECOMMUNICATIONS ENGINEER

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PROFILE

FPGA and ASIC engineer specializing in High-Level Synthesis (HLS) for high-performance wireless communication systems. Experienced in architecting reusable hardware IP in C++ and RTL, optimizing designs for latency, throughput, timing, and power, and bridging digital design, embedded software, and verification across telecom and aerospace.

SKILLS

Programming & HDL	C++, C, SystemVerilog, Verilog, VHDL, Python, MATLAB, Bash.
HLS & FPGA Flow	Vitis HLS, Catapult HLS, Vivado, Vitis, Libero SoC, synthesis, place & route, timing closure.
FPGA & SoC	Zynq-7000 SoC, Zynq UltraScale+ RFSoc/MPSoc, Kintex UltraScale, Microsemi ProASIC3, PetaLinux.
Verification	UVM, Questa Sim, VCS, hardware-in-the-loop, golden-reference models.
DSP & Telecom	DSP, Statistical Signal Processing, Beamforming, Software Defined Radio, GNU Radio, Machine Learning, Optical/RF/Wireless/Analog/Digital/Satellite Communications, Fundamentals of Optical Fibers, CST Studio, ADS.
CI/CD & Tooling	Jenkins, Make, GitLab, GitHub, Gerrit, Jira, Doxygen, L ^A T _E X, Markdown, Visual Studio Code.
Networking & Systems	CCNA R&S (Level 2), pfSense, MikroTik, OpenVPN, WireGuard, Tailscale, Nginx, Keycloak, Cloudflare, Proxmox, Linux (Arch, Ubuntu Server, RHEL), Windows, Cisco IOS, RouterOS.
Home Lab	Home Assistant, TrueNAS, Zigbee2MQTT, Matter, AdGuard.
Languages	English (C1), Spanish (Native), Dutch (A1).

DIGITAL DESIGN & EMBEDDED ENGINEERING EXPERIENCE

🏢 **Ericsson**
MINI-LINK

Dec 2022 - Now
Gothenburg, Sweden (Remote)

FPGA and ASIC Developer Consultant via TMC Electronics focused on developing high-capacity wireless communication systems for the MINI-LINK product family.

- Applied the High-Level Synthesis (HLS) paradigm to improve the quality of results (QoR) of existing IP cores, strictly meeting latency, throughput, timing, and power constraints while accelerating design and verification cycles.
- Architected a C++ abstraction layer enabling unified source code deployment for over 80 distinct hardware IP cores across both Vitis HLS and Catapult HLS, allowing seamless exploration of different hardware targets.
- Developed C++ software drivers, device firmware, and low-level handlers to enable CPU-to-FPGA communication and hardware IP core management.
- Developed functional MATLAB acceptance tests for sub-systems and individual IP cores to validate algorithm behavior against floating-point and fixed-point golden references.
- Designed UVM test cases and hardware-in-the-loop (HIL) tests to verify sub-chip and full-chip functionality on physical hardware under real-world operating constraints.
- Improved the CI/CD pipeline by automating nightly and hourly hardware regression tests in Jenkins (Make, Python, MATLAB, Bash), spanning hundreds of verification cycles per day, and debugging regression errors to maintain integration stability.
- Implemented synthesis and routing strategies to achieve timing closure across highly constrained FPGA designs targeting 250 MHz clock domains.

Digital Design Engineer Consultant via Emtech S.A., member of the team responsible for the development of the BlueWalker 3's "Micron" chip architecture. The BlueWalker 3 was AST's first prototype satellite with the capability of direct-to-smartphone connectivity.

- Designed and refactored communication and control logic for the Micron sub-chips, including digital and analog front ends, LTE systems, and beamforming components.
- Implemented robust design-for-testability (DFT) IP cores to allow remote real-time debugging of the Micron's subsystems.
- Implemented automated on-target verification scripts using Python to command lab instruments and configure the device under test.

 Skyloom Global Corp.
V'ger

Oct 2020 - Oct 2021
Oakland, USA (Remote)

FPGA Design Engineer Consultant via Emtech S.A., part of the team responsible for the development of the V'ger Optical Communications Terminal for inter-satellite optical links.

- Developed Q/SPI, UART, and I2C interfaces to control inter-chip communications and optical peripherals, such as EDFAs and pointing systems.
- Implemented AXI and Wishbone bus bridges to handle communications between the CPU, FPGA, and various board-connected peripherals.
- Created and executed acceptance tests for the electronic boards driving the V'ger system.

EDUCATION

 Instituto Balseiro (UNCuyo)

Jul 2017 - Dec 2020

Telecommunications Engineering (Master's Equivalent)

San Carlos de Bariloche, Argentina

- **Structure:** The concluding 3.5 years of a 5.5-year integrated program (Bologna 300+ ECTS equivalent); admission to this highly selective institute required prior completion of a 2-year university foundational program.
- **Honors:** Awarded a highly competitive, full merit-based CNEA scholarship to complete advanced Master's-level specialization.
- **Final Project:** *Digital beamformer for the reception of satellite signals* (Aug 2019 - Dec 2020) @ Telecommunications Eng. Dept., Centro Atómico Bariloche: Implemented digital beamforming for antenna-array signal reception using the MUSIC and ESPRIT direction-of-arrival algorithms, plus SVM-based signal count estimation. Designed the system architecture for a CIAA-ACC board, partitioning functionality between the FPGA and microprocessor, and validated the complete GNU Radio implementation through Python simulations.

 Facultad de Cs. Exactas, Ingeniería y Agrimensura (UNR)
Electronic Engineering

Feb 2009 - Jul 2017

Rosario, Argentina

- 5-year program, three years completed.

HLS OUTREACH & KNOWLEDGE SHARING

Talks delivered on the HLS paradigm, sharing practical insights, examples, and demonstrations:

- **Dec 2024 – Experiences with HLS @ IMEC:** Presented to IMEC's DSP team on HLS benefits and practical applications in their projects.
- **Apr 2024 – Introduction to HLS @ Instituto Balseiro:** Taught a four-hour class on the HLS paradigm and workflow to Telecommunications Engineering students.
- **Sep 2023 – HLS Pizza Session @ TMC:** Introduction to HLS for 40 participants, including an FFT demo in Vitis HLS.

ADDITIONAL COURSES AND CERTIFICATIONS

Core-Vision

Frank de Bont

- **Essential DSP Implementation Techniques for Xilinx FPGAs (Sep 2025):** A 2-day course on implementing DSP algorithms on Xilinx FPGAs: fixed-point arithmetic, filter design, FFTs, and resource optimization.
- **High-Level Synthesis with the Vitis HLS tool (Dec 2022):** A 2-day overview of the HLS paradigm: C/C++ coding for hardware, optimization techniques, and integration with the Vitis toolchain.

IEEE Boston

Daniel Boschen

- **DSP For Software Radio (Jun 2024 - Jul 2024):** A 5-week online course on DSP fundamentals for software-defined radio: sampling theory, filtering, modulation and demodulation, carrier and timing recovery, quadrature error correction, and equalization.

ADDITIONAL EXPERIENCE

Emtech S.A.

IT & Network Administrator

Oct 2020 - Nov 2022

Rosario, Argentina

- Architected and maintained the internal IT infrastructure, including routers, firewalls, VPNs, GitLab servers, automated backups, and Google Workspace domains.

Centro de Innovación Tecnológica

Instructor

Mar 2016 - Jul 2017

Rosario, Argentina

- IT course instructor at Fundación Libertad: *Cloud Management with GSuite* (80 h, Empleartec Program), *Cisco IT Essentials* (80 h), and *Microsoft Excel, Advanced Level* (12 h, 6 editions).

Consulado General de Chile en Rosario

IT Support Manager

Sep 2013 - Jul 2017

Rosario, Argentina

- Managed a Help Desk team of 7 and acted as technical liaison between the Chilean government and the consulate for new IT deployments.
- Implemented and administered the network infrastructure; advanced hardware and software troubleshooting.

Fundación Libertad

IT Support Manager

Aug 2012 - Jul 2017

Rosario, Argentina

- Managed Help Desk operations for 30 employees: GSuite administration, external web hosting, and user accounts and security policies via Active Directory on Windows Server 2008.

VOLUNTEER EXPERIENCE

TMC Works Council (OR)

IT & Communication Committees Member

Oct 2023 - May 2026

Eindhoven, The Netherlands

- Advised the company's IT department on internal projects from an employee and adoption perspective.
- Improved Works Council communication by updating the intranet and sending quarterly newsletters.

Centro de Estudiantes del Instituto Balseiro

IT Secretary

Apr 2019 - Dec 2020

San Carlos de Bariloche, Argentina

- Elected by vote in 2019 and 2020. Provided Help Desk support and VPN solutions for ≈ 200 students, administered file and web servers, and managed IT purchases and vendor coordination.